

Vibrational Wiring Deduction Transformer (VDT):

A Spectral-PPCA Generative Framework for
Graph-Structured Latent Spaces and Spectral Associative Memory

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Abstract

We present the *Vibrational Wiring Deduction Transformer* (VDT), a fully Bayesian generative model that unifies vibrational graph dynamics, probabilistic PCA in a Laplacian eigenbasis (Spectral-PPCA), and spectral associative memory into a single trainable architecture. The predecessor Wiring Autoencoder (WAE) used a hard high-frequency energy penalty (J_{freq}) and an isotropic Gaussian latent prior. VDT replaces both with principled variational KL terms: (i) a spectral-basis KL over the loading matrix $\mathbf{S}$ weighted by Laplacian eigenvalues, implementing a *spectral Occam's razor*; (ii) a closed-form Gamma/Exponential KL over per-mode weights ω , providing soft τ -mode band-limiting; and (iii) the standard isotropic Gaussian latent KL retained for stability (the Laplacian-precision latent KL is preserved as a design option and ablation, but is not included in the default three-term ELBO following architectural review in PR #35). After training, a *spectral artefact* $\mathcal{A}(\mathcal{I})$ is extracted and used to initialise a downstream transformer with pre-built spectral associative memory. The two-phase (offline training, online inference) architecture separates spectral learning from deployment, with frozen eigenpairs eliminating all runtime $O(N^3)$ eigendecompositions during training. We detail the complete module design, three-term ELBO, stability analysis, benchmark metrics, and associative memory construction.

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1 Introduction

Graph-structured latent variable models occupy an increasingly important niche between classical variational autoencoders (VAEs) and geometric deep learning: they can encode both the *content* of nodes and the *relational structure* that organises them. The Wiring Autoencoder [?] recognised this, using a differentiable graph Laplacian $L(z)$ as an intermediate latent object whose spectral properties are regularised by a hard frequency penalty $J_{\text{freq}} = \sum_{j>k} \lambda_j(L(z))$. While effective in practice, this penalty is a heuristic rather than a proper probabilistic prior, which makes it difficult to:

1. compare competing ArrowSpace indices $\mathcal{I}_1, \mathcal{I}_2$ via principled model selection;
2. interpret the learned spectral structure as a posterior over loading matrices $\mathbf{W}$;
3. extract a reusable *spectral artefact* whose keys are approximately orthonormal and whose values are grounded in the data manifold.

The VDT addresses all three limitations by embedding the Wiring Autoencoder within the Spectral-PPCA framework [Moriondo, 2026a], a Bayesian generative model in which the loading matrix $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$ lives explicitly in the Laplacian eigenbasis and is endowed with eigenvalue-weighted shrinkage priors. The VDT encoder backbone implements a discrete damped wave equation in the graph spectral domain, connecting the probabilistic latent space to the physical intuition of vibrational modes in string and spring systems.

1.1 Contributions

- A three-term ELBO replacing J_{freq} and preserving the isotropic latent prior (Section 4).
- A **SpectralLoadingDecoder** that maps latent codes to eigenbasis-factorised loading matrices (Section 5).
- A **VibrationalEncoder** (VDT recurrence) with **ModeWeightHead** producing variational Gamma parameters (Section 5).
- A **SpectralAssociativeMemory** module that pre-builds a Hopfield memory from the spectral artefact (Section 6).
- A two-phase offline/online architecture eliminating runtime eigensolves during training (Section 7).
- An eight-metric evaluation suite including ELBO Bayes factors for ArrowSpace index selection (Section 9).

2 Background

2.1 Probabilistic PCA

Standard PPCA [Tipping and Bishop, 1999] defines a linear-Gaussian generative model:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_q), \quad \mathbf{x} \mid \mathbf{z} \sim \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I}_d), \quad (1)$$

producing marginal $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^\top + \sigma^2 \mathbf{I}_d)$. Maximum-likelihood estimation recovers the PCA loading matrix in the column space of the q largest singular vectors of the data matrix. The fully Bayesian extension places priors over $\mathbf{W}$, $\mathbf{z}$, and σ^2 , enabling model selection over the latent dimensionality q via the marginal likelihood [Minka, 2001].

2.2 Graph Laplacian Spectral Theory

Let $G = (V, E, w)$ be a weighted graph with node set $|V| = N$. The normalised graph Laplacian is $L = D^{-1/2}(D - W)D^{-1/2}$ where $D_{ii} = \sum_j w_{ij}$. Eigendecomposition $L = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ yields eigenvectors $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N) \in \mathbb{R}^{N \times N}$ and non-negative eigenvalues $0 \leq \lambda_1 \leq \dots \leq \lambda_N$. Low-frequency eigenvectors (small λ_k) are smooth over G ; high-frequency ones (large λ_k) oscillate rapidly across edges [Chung, 1997].

Given ArrowSpace index $\mathcal{I}$, we denote the frozen q -truncated eigendecomposition

$$L(\mathcal{I}) = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top, \quad \mathbf{U}_q = \mathbf{U}_{:,1:q} \in \mathbb{R}^{N \times q}, \quad \mathbf{\Lambda}_q = \text{diag}(\lambda_1, \dots, \lambda_q). \quad (2)$$

These eigenpairs are computed once offline and used as frozen constants throughout training.

2.3 Vibrational Dynamics and Spectral Graph Waves

The VDT encoder draws its physical motivation from Rayleigh’s Theory of Sound [?]: the eigenmodes of a vibrating string or spring system are exactly the eigenvectors of the system’s stiffness matrix, which in the discrete graph setting is the normalised Laplacian L . A positive amplitude on mode k corresponds to a string displacement in the direction $\mathbf{u}_k$; a negative amplitude corresponds to the opposing displacement. This is the physical counterpart to the signed (positive/negative) latent amplitudes that the VDT encoder learns.

The signed density matrix $\varrho_t = \varrho_t^+ - \varrho_t^-$ maintained by the VDT recurrence (Section 5.2) is precisely a decomposition of the current state into constructive and destructive vibrational contributions. The low-entropy state of this system corresponds to coherent oscillation (the string vibrating in a definite mode); the high-entropy state corresponds to rest (the wave function collapsed, all modes equally weighted). This mirrors Brownian motion, where the low-entropy state is the non-heated stationary system and the high-entropy state is maximal thermal agitation.

2.4 Wiring Autoencoder (WAE) — Predecessor

WAE [?] trains a differentiable Laplacian $L(\mathbf{z})$ whose topology is fixed (kNN + RBF graph) but whose edge weights are learned by a mixture-of-experts decoder $W_{\text{dec}} : \mathbf{z} \mapsto \delta_e$. The training objective is

$$\mathcal{L}_{\text{WAE}} = \mathbb{E}_q[\log p(\mathbf{x} \mid \mathbf{z})] - \beta \cdot \text{KL}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) - \alpha \cdot J_{\text{freq}}(L(\mathbf{z})), \quad (3)$$

where $J_{\text{freq}} = \sum_{j>k} \lambda_j(L(\mathbf{z}))$ penalises high-frequency wiring. The reconstruction is provided by **TauModeDiffusion**: the heat kernel $K_\tau = \mathbf{U}_k \exp(-t\mathbf{\Lambda}_k)\mathbf{U}_k^\top$ applied to an embedding table $\mathbf{E}$, with learnable diffusion time t .

Three limitations of WAE motivate VDT: (a) J_{freq} is a deterministic penalty with no prior interpretation; (b) the latent prior is isotropic and ignores graph geometry; (c) there is no principled post-training artefact for downstream use.

3 Spectral-PPCA Generative Framework

Following Moriondo [2026a], we define the Spectral-PPCA prior by replacing the standard PPCA loading matrix with one that lives in the Laplacian eigenbasis.

3.1 Three Structural Priors

Prior 1: Spectral-basis loading. Parametrise $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$ where $\mathbf{U}_{1:q} \in \mathbb{R}^{d \times q}$ are the q lowest-frequency eigenvectors of $L(\mathcal{I})$ (frozen) and $\mathbf{S} \in \mathbb{R}^{q \times q}$ lives in the spectral basis. Place an eigenvalue-weighted Gaussian shrinkage prior on $\mathbf{S}$:

$$p(\mathbf{S} \mid \mathcal{I}) \propto \exp\left(-\frac{\lambda_s}{2} \text{tr}(\mathbf{S}^\top \mathbf{\Lambda}_q \mathbf{S})\right). \quad (4)$$

This implements a *spectral Occam’s razor*: high-frequency components (large λ_k) are penalised exponentially more strongly than low-frequency ones, pulling the loading matrix toward smooth directions of the index graph.

Prior 2: τ -mode frequency prior. Let $\omega_k > 0$ be a weight on the k -th mode. The τ -mode prior is a product of exponentials with rates that scale with Laplacian eigenvalues:

$$p(\omega_k \mid \tau, \lambda_k) = \text{Exp}(\tau \lambda_k), \quad k = 1, \dots, q. \quad (5)$$

Small λ_k (low-frequency / low-entropy vibrational modes) receive weak penalisation; large λ_k receive strong penalisation, naturally selecting the effective spectral bandwidth. The loading matrix is $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$.

Prior 3 (design option): Laplacian-precision latent prior. For a richer latent geometry one can replace the isotropic prior with a Laplacian-precision prior:

$$p(\mathbf{z} \mid \mathcal{I}) = \mathcal{N}(\mathbf{0}, (\mathbf{I} + \beta L_s)^{-1}), \quad (6)$$

where L_s is a sample-graph Laplacian built from the current batch of latent means (stop-gradient). The resulting KL expands to

$$\text{KL}(q(\mathbf{z}) \parallel p_{\text{Lap}}(\mathbf{z})) = \frac{1}{2} \left[\text{tr}(\Sigma_z (\mathbf{I} + \beta L_s)) + \boldsymbol{\mu}_z^\top (\mathbf{I} + \beta L_s) \boldsymbol{\mu}_z - d - \log \det \Sigma_z + \log \det (\mathbf{I} + \beta L_s)^{-1} \right], \quad (7)$$

where the cross-term $\boldsymbol{\mu}_z^\top L_s \boldsymbol{\mu}_z$ is the graph Dirichlet energy of the latent means. *This prior is preserved in the VDT codebase as an ablation option (use_isotropic_kl=False in VibrationalEncoder)* but is **not** included in the default three-term ELBO following architectural review in PR #35.

4 VDT: Three-Term ELBO

4.1 Variational Family

We adopt a mean-field amortised variational family:

$$q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_z, \text{diag}(\exp(\mathbf{s}_z))), \quad (8)$$

$$q_\phi(\mathbf{S} \mid \mathbf{x}) = \mathcal{N}(M_S, \text{diag}(\exp(\mathbf{v}_S))), \quad (9)$$

$$q_\phi(\omega_k \mid \mathbf{x}) = \text{Gamma}(a_k, b_k), \quad (10)$$

where all parameters are outputs of the `VibrationalEncoder` network. The Gamma distribution is reparametrisable via the implicit reparametrisation trick [Figurnov et al., 2018]; in practice we parametrise through $(\log a_k, \log b_k)$ from a `ModeWeightHead` linear layer.

4.2 ELBO Decomposition

The default VDT training objective is the following three-term ELBO (minimisation form, as implemented in `vdt/model.py`):

$$\mathcal{L}_{\text{VDT}} = \underbrace{\mathbb{E}_q[\|\mathbf{x} - \hat{\mathbf{x}}\|^2 / 2\sigma^2]}_{\text{reconstruction}} + \underbrace{\text{KL}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{isotropic latent KL (kl_z)}} + \underbrace{\text{KL}(q(\mathbf{S}) \parallel p(\mathbf{S} \mid \mathcal{I}))}_{\text{spectral basis KL (kl_S)}} + \underbrace{\text{KL}(q(\boldsymbol{\omega}) \parallel p(\boldsymbol{\omega} \mid \tau, \boldsymbol{\Lambda}))}_{\tau\text{-mode KL (kl_tau)}} \quad (11)$$

All four terms are non-negative; the total loss is minimised (ELBO is maximised by negating) during gradient descent.

4.3 Closed-Form KL Terms

Isotropic latent KL.

$$\text{KL}(q(\mathbf{z})\|\mathcal{N}(\mathbf{0}, \mathbf{I})) = -\frac{1}{2} \sum_{i=1}^{d_z} (1 + s_{z,i} - \mu_{z,i}^2 - e^{s_{z,i}}). \quad (12)$$

Spectral-basis KL. For the entry (k, j) of $\mathbf{S}$, with precision $\lambda_s \lambda_k$ (eq. (4)):

$$\text{KL}_{kj} = \frac{1}{2} [\lambda_s \lambda_k (v_{S,kj} + S_{kj}^2) - v_{S,kj} - 1 - \log(\lambda_s \lambda_k)], \quad (13)$$

where $v_{S,kj} = \exp(\log \text{var}_{S,kj})$. In the current implementation the log-variance is approximated as $\log \text{var}_{S,kj} = \log(S_{kj}^2 + \epsilon)$, deriving uncertainty from the magnitude of the mean — a simplification that avoids adding a separate posterior variance head to `SpectralLoadingDecoder`. Future work should add a dedicated variance head for a proper amortised posterior.

τ -mode KL. For mode k with $q(\omega_k) = \text{Gamma}(a_k, b_k)$ and $p(\omega_k) = \text{Gamma}(1, \tau \lambda_k)$, the closed-form KL is:

$$\text{KL}(\text{Gamma}(a, b)\|\text{Exp}(r)) = \log b - \log r + \ln \Gamma(a) + (1 - a)\psi(a) + \frac{ab}{r}, \quad r = \tau \lambda_k, \quad (14)$$

where ψ is the digamma function. This is summed over all q modes and averaged over the batch.

4.4 Comparison with WAE Predecessor

Table 1: ELBO comparison: WAE (predecessor) vs VDT.

Term	WAE	VDT	Note
Reconstruction	$-\ \mathbf{x} - \hat{\mathbf{x}}\ ^2/2\sigma^2$	Same	σ learnable
Latent KL	$\text{KL}(q(\mathbf{z})\ \mathcal{N}(\mathbf{0}, \mathbf{I}))$	Same (isotropic)	PR #35 decision
Spectral basis KL	—	$\text{KL}(q(\mathbf{S})\ p(\mathbf{S} \mathcal{I}))$	eigenvalue-weighted Gaussian
τ -mode term	αJ_{freq} (hard)	$\text{KL}(q(\boldsymbol{\omega})\ p(\boldsymbol{\omega} \tau, \mathbf{A}))$	variational Gamma prior
Lap. precision latent KL	—	<i>design option</i>	ablation only

5 Model Architecture

5.1 Data Flow

Figure 1 illustrates the end-to-end data flow.

5.2 VibrationalEncoder

`VibrationalEncoder` (`vdt/encoder.py`) replaces the WAE MLP encoder with a VDT (Vibrational Discrete Transformer) recurrence stack:

1. **Input projection:** linear layer mapping $(B, D) \rightarrow (B, N \cdot d)$, reshaped to (B, N, d) .
2. **VibrationalStateBlocks:** K recurrence layers operating on (B, N, d) state matrices using the feature-space Laplacian L_f . The recurrence implements a discrete damped wave equation in the graph spectral domain: the state evolves as $\mathbf{h}_{t+1} = (2\mathbf{I} - \alpha L_f)\mathbf{h}_t - \mathbf{h}_{t-1}$, where α is a learnable damping coefficient. This is a finite-difference discretisation of the wave equation $\ddot{\mathbf{u}} + \alpha L_f \mathbf{u} = 0$.

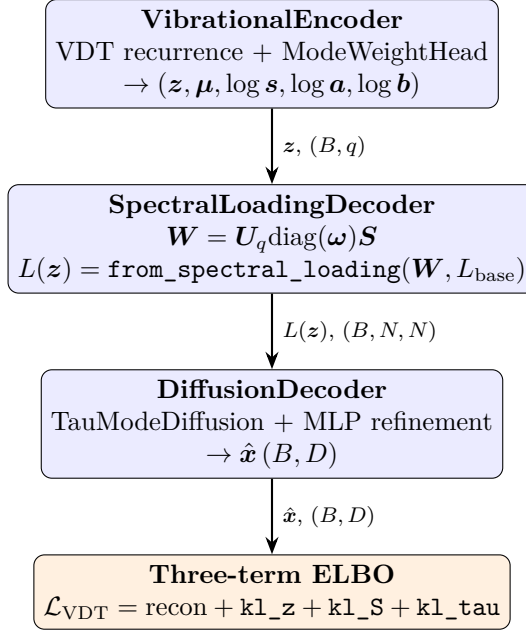


Figure 1: VDT data flow. The eigenpairs $(\mathbf{U}_q, \mathbf{\Lambda}_q)$ of $L(\mathcal{I})$ are frozen constants passed to both encoder and decoder; no Laplacian is computed at runtime.

3. **Modal projection:** mean-pool output over the node dimension after projecting through eigenvectors $\mathbf{v} \in \mathbb{R}^d$.
4. **Readout heads:** separate linear layers producing $\boldsymbol{\mu}, \log \mathbf{s}$ (for $q(\mathbf{z})$) and **ModeWeightHead** producing $(\log \mathbf{a}, \log \mathbf{b})$ for $q(\boldsymbol{\omega})$.

The optional λ -fingerprint (ArrowSpace spectral histogram of $L(\mathcal{I})$) can be concatenated to $\mathbf{x}$ before the input projection (`use_lambda_features=True`).

ModeWeightHead. A single linear layer `Linear(feats_dim, 2q)` splits into $(\log \mathbf{a}, \log \mathbf{b}) \in \mathbb{R}^q \times \mathbb{R}^q$. The Gamma parameters are obtained as $a_k = e^{\log a_k} > 0$, $b_k = e^{\log b_k} > 0$. The mode-weight KL (14) is computed from these in `vdtspectral.py:tau_mode_kl`.

Signed density matrix. During the VDT recurrence each block maintains a signed density matrix $\varrho_t = \varrho_t^+ - \varrho_t^-$, decomposing the state into constructive and destructive vibrational contributions (positive and negative amplitudes). This object is the discrete analogue of the quantum density matrix when amplitudes are real-valued rather than complex. It is used as a stability diagnostic and, optionally, as the seed for the spectral artefact in reasoning-focused deployments (Option 6, Section 10).

5.3 SpectralLoadingDecoder

`SpectralLoadingDecoder` (`vdts/wiring_decoder.py`) is the default VDT wiring decoder. The forward pass computes:

$$\mathbf{S} = \mathbf{S_net}(\mathbf{z}).\text{view}(B, q, q), \quad (15)$$

$$\boldsymbol{\omega} = \exp(\boldsymbol{\omega_net}(\mathbf{z})), \quad (16)$$

$$\mathbf{W} = \mathbf{U}_q.\text{unsqueeze}(0) @ (\boldsymbol{\omega}.\text{unsqueeze}(-1) * \mathbf{S}), \quad (17)$$

$$L(\mathbf{z}) = \text{DifferentiableLaplacian.from_spectral_loading}(\mathbf{W}, L_{\text{base}}). \quad (18)$$

Edge weights in (18) are synthesised as $w_{ij} = w_{ij}^{\text{base}} \cdot \sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2)$ where $\mathbf{W}_i$ is the i -th row of $\mathbf{W}$ and σ is the logistic sigmoid. This path is fully differentiable from $L(\mathbf{z})$ back to $\mathbf{z}$ through $\mathbf{W}$.

Initialisation. `S_net.weight` is initialised so the initial $\mathbf{S}_b \approx \mathbf{I}_q$ (near-identity); `omega_net` is initialised near zero so initial $\omega_k \approx 1$ for all modes. This gives a warm-start close to the eigenbasis reparametrisation $\mathbf{W} \approx \mathbf{U}_q$.

Config dispatch. The legacy WAE `WiringDecoder` (mixture-of-experts) is retained as a fallback: `decoder_type: spectral` selects `SpectralLoadingDecoder`; `decoder_type: mixture_of_experts` selects `WiringDecoder`.

5.4 DiffusionDecoder

`DiffusionDecoder` is retained from WAE. It wraps `TauModeDiffusion`: the truncated heat kernel

$$K_\tau = \mathbf{U}_k \text{diag}(e^{-t\Lambda_k}) \mathbf{U}_k^\top, \quad (19)$$

with learnable log-diffusion-time $\log t$. A small MLP refinement residually corrects the output. Reconstruction loss is Gaussian NLL: $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 / (2\sigma^2)$.

The heat kernel (19) has a direct vibrational interpretation: $e^{-t\lambda_k}$ is the amplitude decay factor for mode k after time t , exactly as in the damped wave equation. Modes with large λ_k (high-frequency, high-entropy) decay faster; low-frequency modes persist.

5.5 VibrationalDeductionTransformer (Top-level Module)

`VibrationalDeductionTransformer` (`vdt/model.py`) assembles all three modules. Its `forward()` method returns a dict with exactly nine keys: `{loss, recon, kl_z, kl_S, kl_tau, x_hat, z, mu, log_var}`.

Model factory. `from_config(cfg, E)` dispatches on `cfg['model']['version']`: version 1 builds the legacy `WiringAutoencoder`; version 2 builds `VibrationalDeductionTransformer`.

6 Spectral Artefact and Associative Memory

6.1 Spectral Artefact

After training to convergence, the spectral artefact $\mathcal{A}(\mathcal{I})$ is extracted by evaluating the model at the prior mean $\mathbf{z} = \mathbf{0}$:

$$\mathcal{A}(\mathcal{I}) = \left\{ \hat{\mathbf{W}} = \mathbf{U}_q \text{diag}(\hat{\boldsymbol{\omega}}) \hat{\mathbf{S}}, \quad \hat{\boldsymbol{\omega}} = (\hat{\omega}_1, \dots, \hat{\omega}_q), \quad S_{\mathcal{I}} = \sum_{k=1}^q \hat{\omega}_k \hat{\mathbf{w}}_k \hat{\mathbf{w}}_k^\top \right\}, \quad (20)$$

where $\hat{\mathbf{w}}_k$ is the k -th column of $\hat{\mathbf{W}}$. Equation (20) uses $d_\theta(\hat{\mathbf{w}}_k) \approx \hat{\mathbf{w}}_k$ (linear decoder approximation for $S_{\mathcal{I}}$); a full nonlinear value function $d_\theta(\hat{\mathbf{w}}_k)$ is left as future work (Section 12).

Table 2: Components of the spectral artefact $\mathcal{A}(\mathcal{I})$.

Component	Symbol	Meaning
Posterior loading basis	$\hat{\mathbf{W}} \in \mathbb{R}^{d \times q}$	Mean loading matrix in Laplacian eigenbasis
Posterior mode weights	$\hat{\boldsymbol{\omega}} \in \mathbb{R}^q$	Per-mode activity under τ -mode prior
Associative memory matrix	$S_{\mathcal{I}} \in \mathbb{R}^{d \times d}$	Outer-product Hopfield key matrix

6.2 SpectralAssociativeMemory

`SpectralAssociativeMemory` (`vdt/spectral_memory.py`) wraps $S_{\mathcal{I}}$ as a pre-built Hopfield / linear associative memory. For query $\mathbf{q} \in \mathbb{R}^{d_k}$, retrieval proceeds via:

$$\mathbf{o} = \text{softmax}\left(\beta_H S_{\mathcal{I}}^{\top} \mathbf{q}\right) S_{\mathcal{I}}, \quad (21)$$

where β_H is the inverse temperature controlling retrieval sharpness. The approximately-orthonormal keys $\hat{\mathbf{w}}_k$ (columns of $\hat{\mathbf{W}}$ are linear combinations of eigenvectors, orthonormal when $\mathbf{S} \approx \mathbf{I}$) maximise retrieval SNR [Ramsauer et al., 2021]: $\text{SNR}^{-1} \approx N_{\text{stored}}/d_k$.

Vibrational interpretation. The columns of $\hat{\mathbf{W}}$ are superpositions of Laplacian eigenmodes: low-frequency modes (small λ_k , high $\hat{\omega}_k$) correspond to the fundamental vibrational tones of the graph, while high-frequency modes correspond to harmonics that are suppressed by the τ -mode prior. The memory matrix $S_{\mathcal{I}}$ thus stores information in the basis of the system’s natural resonances — an associative memory grounded in physical intuition.

Online delta-rule updates. New associations can be written without overwriting spectral structure:

$$S_{\mathcal{I}} \leftarrow S_{\mathcal{I}} + \frac{1}{B}(\mathbf{v} - S_{\mathcal{I}}\mathbf{k})\mathbf{k}^{\top}, \quad (22)$$

where $(\mathbf{k}, \mathbf{v})$ are new key-value pairs.

7 Two-Phase Architecture

VDT cleanly separates spectral learning from online inference.

Algorithm 1 VDT Two-Phase Procedure

Require: ArrowSpace index $\mathcal{I}$, dataset D , hyperparameters

- 1: **// Phase 1 — Offline spectral learning**
 - 2: Compute eigendecomposition: $(\mathbf{U}_q, \mathbf{\Lambda}_q) \leftarrow \text{eigh}(L(\mathcal{I}))$ $\triangleright$ once; $O(N^3)$
 - 3: Freeze $(\mathbf{U}_q, \mathbf{\Lambda}_q)$
 - 4: **for** each training step **do**
 - 5: $\mathbf{z}, \boldsymbol{\mu}, \log \mathbf{s}, \log \mathbf{a}, \log \mathbf{b} \leftarrow \text{VibrationalEncoder}(\mathbf{x})$
 - 6: $\mathbf{W}, \boldsymbol{\omega}, \mathbf{S}, L(\mathbf{z}) \leftarrow \text{SpectralLoadingDecoder}(\mathbf{z}, \mathbf{U}_q, L_{\text{base}})$
 - 7: $\hat{\mathbf{x}} \leftarrow \text{DiffusionDecoder}(L(\mathbf{z}), \mathbf{E})$
 - 8: Compute $\mathcal{L}_{\text{VDT}}$ (eq. (11)) and update ϕ, θ
 - 9: **end for**
 - 10: $\mathcal{A}(\mathcal{I}) \leftarrow \text{extract_spectral_artefact}(\mathbf{U}_q, \mathbf{\Lambda}_q)$
 - 11: $S_{\mathcal{I}} \leftarrow \text{SpectralAssociativeMemory}(\mathcal{A}(\mathcal{I}))$
 - 12: **// Phase 2 — Online inference**
 - 13: Initialise transformer FFN / cross-attention from $S_{\mathcal{I}}$
 - 14: **for** each inference query $\mathbf{q}$ **do**
 - 15: Retrieve via eq. (21); optionally update via eq. (22)
 - 16: **end for**
-

No runtime eigensolves. Because $(\mathbf{U}_q, \mathbf{\Lambda}_q)$ are frozen constants in Phase 1, *no Laplacian is computed or inverted at each training step*. All $O(N^3)$ eigendecomposition cost is paid once in offline preprocessing.

ELBO Bayes factor for index selection. Training VDT on competing indices $\mathcal{I}_1, \mathcal{I}_2$ yields ELBO estimates $\mathcal{L}(\mathcal{I}_1), \mathcal{L}(\mathcal{I}_2)$. The Bayes factor

$$\text{BF}_{12} = \exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2)) \quad (23)$$

provides a principled comparison of index constructions: an index whose spectral geometry explains the data covariance better will achieve a higher ELBO [Lotfi et al., 2022].

8 Stability Analysis

VDT inherits the full stability hierarchy from `docs/v2/04-stability.md`. Two additional conditions apply.

Sample-graph Laplacian feedback. When the Laplacian-precision latent KL (design option) is enabled, L_s is built from the batch of latent means μ_z , creating a feedback loop. Prevention: apply stop-gradient on L_s construction; or use a frozen sample Laplacian refreshed every T_{refresh} epochs.

Spectral-basis KL conditioning. If $\lambda_1 \approx 0$ (Fiedler vector near zero), the prior on the first mode becomes nearly flat, increasing gradient variance. Apply a floor $\lambda_k \leftarrow \max(\lambda_k, \epsilon)$ with $\epsilon = 10^{-3}$ in all KL computations (implemented in `vdt/spectral.py` as `eigvals.clamp(min=1e-6)`).

Mode weight collapse. High-frequency modes may collapse to $\omega_k \rightarrow 0$ if $\tau\lambda_k$ is large. Monitor active mode count $|\{k : \mathbb{E}[\omega_k] > \delta\}|$; enforce a soft floor on the Gamma shape $a_k \geq a_{\min}$.

9 Benchmark Metrics

`evaluate()` in `vdt/metrics.py` computes the eight metrics in Table 3. The ELBO Bayes factor enables ArrowSpace index selection from data alone.

Table 3: VDT benchmark metrics.

Metric	Computation	Measures
Reconstruction MSE	$\mathbb{E}_{\text{test}}[\ \mathbf{x} - \hat{\mathbf{x}}\ ^2]$	Quality of wiring-path reconstruction
KL spectral basis	$\text{KL}(q(\mathbf{S})\ p(\mathbf{S} \mid \mathcal{I}))$	Spectral alignment with index
τ -mode KL	$\text{KL}(q(\boldsymbol{\omega})\ p(\boldsymbol{\omega} \mid \tau, \boldsymbol{\Lambda}))$	Frequency band selection
Active mode count	$\sum_k \mathbf{1}[\mathbb{E}[\omega_k] > 0.01]$	Effective spectral bandwidth
Memory retrieval SNR	d_k/N_{stored} (key orthogonality)	Associative memory quality
ELBO Bayes factor	$\exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2))$	ArrowSpace index comparison
Linear probe accuracy	Logistic regression on frozen μ	Discriminative latent quality
Spectral entropy $H(\Lambda)$	$H(\text{normalised eigenvalues of } L(\mathbf{z}))$	Wiring diversity

10 Algorithm Tracks

The VDT framework supports six implementation tracks. The recommended sequence is $6 \rightarrow 1 \rightarrow 4 \rightarrow 3$.

- **Option 1 — Deterministic AE.** Replace J_{freq} with the soft τ -mode KL; add `SpectralLoadingDecoder` as ablation. Backward-compatible: $\beta \rightarrow 0$ recovers the deterministic limit.

- **Option 3 — Latent diffusion.** VDT provides the Stage 1 encoder; the spectral noise schedule $a_\tau^{(k)} = \exp(-\tau\lambda_k t)$ matches the Λ_q^{-1} latent prior.
- **Option 4 — Variational Laplace AE.** VDT is the full MC-ELBO version; Laplace covariance $(H_{\text{recon}} + \Lambda_m)^{-1}$ is the deterministic approximation to the spectral-precision posterior.
- **Option 6 — Classifier/Reasoner.** The signed density matrix $\varrho_t = \varrho_t^+ - \varrho_t^-$ from the `VibrationalStateBlock` backbone can be used to initialise $S_{\mathcal{I}}$, grounding associative memory in reasoning-time spectral structure.

11 Relationship to Prior Work

Graph VAEs. Kipf and Welling [2016] encode graph structure into a VAE but do not impose Laplacian spectral priors on latent codes or loading matrices. VDT makes the spectral structure explicit and variational.

Spectral graph convolution. Graph spectral methods [Defferrard et al., 2016] use Laplacian eigenvectors as filter bases; VDT places this basis *in the generative model* as the loading matrix parametrisation.

Laplacian priors. Rao et al. [2008] and Slawski and Hein [2015] use Laplacian quadratic-form priors for Bayesian variable selection; VDT applies this to both the loading matrix $\mathbf{S}$ and (optionally) the latent code $\mathbf{z}$.

Hopfield / associative memory. Ramsauer et al. [2021] extended Hopfield networks to continuous attractors; Geva et al. [2021] showed that transformer FFN layers are key-value memories. VDT unifies these by seeding the memory with spectral artefacts derived from vibrational graph dynamics.

Vibrational mechanics. The `VibrationalStateBlock` recurrence is a direct discretisation of the damped wave equation, connecting the architecture to Rayleigh’s Theory of Sound [?]. The positive/negative amplitude decomposition $(\varrho_t^+, \varrho_t^-)$ is a real-valued analogue of the complex amplitude representation in standard quantum mechanics, as discussed in ? Chapter 9.

12 Discussion and Open Problems

Posterior variance head for $\mathbf{S}$. The current implementation approximates $\log \text{var}_{S,kj} \approx \log(S_{kj}^2 + \epsilon)$. Adding a dedicated variance head in `SpectralLoadingDecoder` would give a proper amortised Gaussian posterior for $\mathbf{S}$ and is a prioritised open task.

Laplacian-precision latent KL. PR #35 removed this term from the default ELBO after finding that the feedback loop between μ_z and the dynamically-constructed L_s required careful stop-gradient discipline and added training instability. A frozen sample Laplacian refreshed every T epochs may resolve this; experiments are ongoing.

Nonlinear value function. The current spectral artefact uses a linear decoder approximation $d_\theta(\hat{\mathbf{w}}_k) \approx \hat{\mathbf{w}}_k$ for constructing $S_{\mathcal{I}}$. Using the full `DiffusionDecoder` response at each spectral direction would produce a proper key-value Hopfield structure and is left as future work.

Iterative eigensolver. For large graphs ($N \gg 10^3$), truncated LOBPCG or ARPACK can replace full `eigh` in the offline phase, reducing the preprocessing cost from $O(N^3)$ to $O(N \cdot q)$.

Complex-number-free quantum analogy. A core motivation for the signed density matrix $\varrho_t = \varrho_t^+ - \varrho_t^-$ is to represent quantum-like superposition without complex numbers, following the observation in ? that negative probability amplitudes already capture the essential interference phenomena of quantum mechanics. The graph Laplacian in feature space serves as the minimal structure that admits this representation: its eigenmodes carry the role of quantum basis states, and the signed weights $\omega_k^+ - \omega_k^-$ play the role of real-valued amplitudes. Formalising this correspondence is an open theoretical question.

13 Conclusion

The (VDT) reframes the Wiring Autoencoder as a principled Bayesian VAE in which the latent space and loading matrix are explicitly shaped by Laplacian spectral geometry provided by an ArrowSpace index. The three-term ELBO (eq. (11)) replaces the hard J_{freq} penalty with interpretable variational KL terms grounded in vibrational physics; the two-phase offline/online architecture eliminates runtime eigensolves; and the spectral artefact $\mathcal{A}(\mathcal{I})$ provides a principled initialiser for transformer associative memory whose keys are the natural resonances of the data graph. ELBO Bayes factors make ArrowSpace index selection a first-class modelling decision. The signed density matrix $\varrho_t = \varrho_t^+ - \varrho_t^-$ maintained throughout the `VibrationalStateBlock` recurrence establishes a real-valued analogue of quantum amplitude representation, connecting the architecture to both Rayleigh’s theory of vibrating systems and the foundational question of whether negative probability amplitudes suffice to recover quantum-like behaviour without complex numbers.

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